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The Editors
European Journal of Operational Research

Dear Editors,

Please consider the enclosed manuscript, *"Optimal-stopping recourse for vehicle routing with simultaneous pickup, delivery, and stochastic demands,"* for publication in the *European Journal of Operational Research*.

The paper addresses a routing question in the journal's core tradition of stochastic vehicle routing with recourse: when a delivery vehicle that also collects pickups is filling up mid-route, at which stop should the operator intervene—hand the remainder of the route to a standby vehicle, detour to the depot to unload, or keep driving? We show that this execution-stage question is a finite-horizon optimal-stopping problem and solve it with a policy, BATON, that prices every recourse action from the route's real geometry and intervenes exactly when continuing becomes dearer than the cheapest lever. The policy has no tunable threshold; we prove that no fixed threshold can reproduce the optimal intervention boundary, and that the endpoint-based risk labels used by prior work are structurally blind to mid-route load peaks.

The computational study is, to our knowledge, the broadest comparison of execution-stage recourse policies in this problem class: eleven policies—including published rule-based restocking triggers, rollout, depot-return restocking, a plug-in dynamic program, and a GPU-trained reinforcement-learning baseline—evaluated on identical plans from six planning regimes (three of them published robust planners), on classical benchmarks and on new instances built from OpenStreetMap road networks with customers at real retail locations in five cities. All underlying routing plans are certified against mixed-integer programming bounds, and every table and figure in the paper regenerates mechanically from the result files.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. The authors—Quang-Vinh Dang and Hoang-Viet Vu (British University Vietnam) and Phuc-Son Nguyen (University of Economics Ho Chi Minh City)—have no conflicts of interest to declare, and the work received no external funding; a separate declaration of competing interests and a CRediT authorship statement accompany this submission. The code and data repository will be shared with the editorial office on request during review and made public upon acceptance.

Thank you for your consideration.

Sincerely, on behalf of all authors,

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